

FACIAL EXPRESSION RECOGNITION (FER)

CUSTOM CNN VS RESNET-18 ON FER2013

Group Beta - Aprenentatge Profund 2025-26

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PRESENTATION OUTLINE

- Introduction
- State of the art
- Methodology
- Experiments
- Results
- Discussion
- Conclusion
- References

1-INTRODUCTION

What's Facial Expression Recognition ?

task of automatically classifying human emotions from facial images using deep learning



Real-world applications:

1. Human-computer interaction & affective computing
2. Mental health monitoring & therapy assistance
3. Automotive safety (driver drowsiness detection)
4. Market research & user experience studies

2- STATE OF THE ART

FER2013 Benchmark — Selected Methods

Method	Year	Test Acc.
Human-level benchmark	—	~65%
VGG-Face + SVM (Mollahosseini et al.)	2016	75.2%
ResMaskNet (Phạm et al.)	2021	76.8%
EfficientNet-B4 + attention	2022	~73%
ResNet-18 (ours, fine-tuned)	2025	63.4%
Custom CNN (ours, scratch)	2025	57.1%

} Our models are positioned as educational implementations — competitive for single-GPU fine-tuning on a noisy benchmark.

3- METHODOLOGY

A- DATA ANALYSIS

35887

TOTAL SAMPLES

28709

TRAINING IMAGES

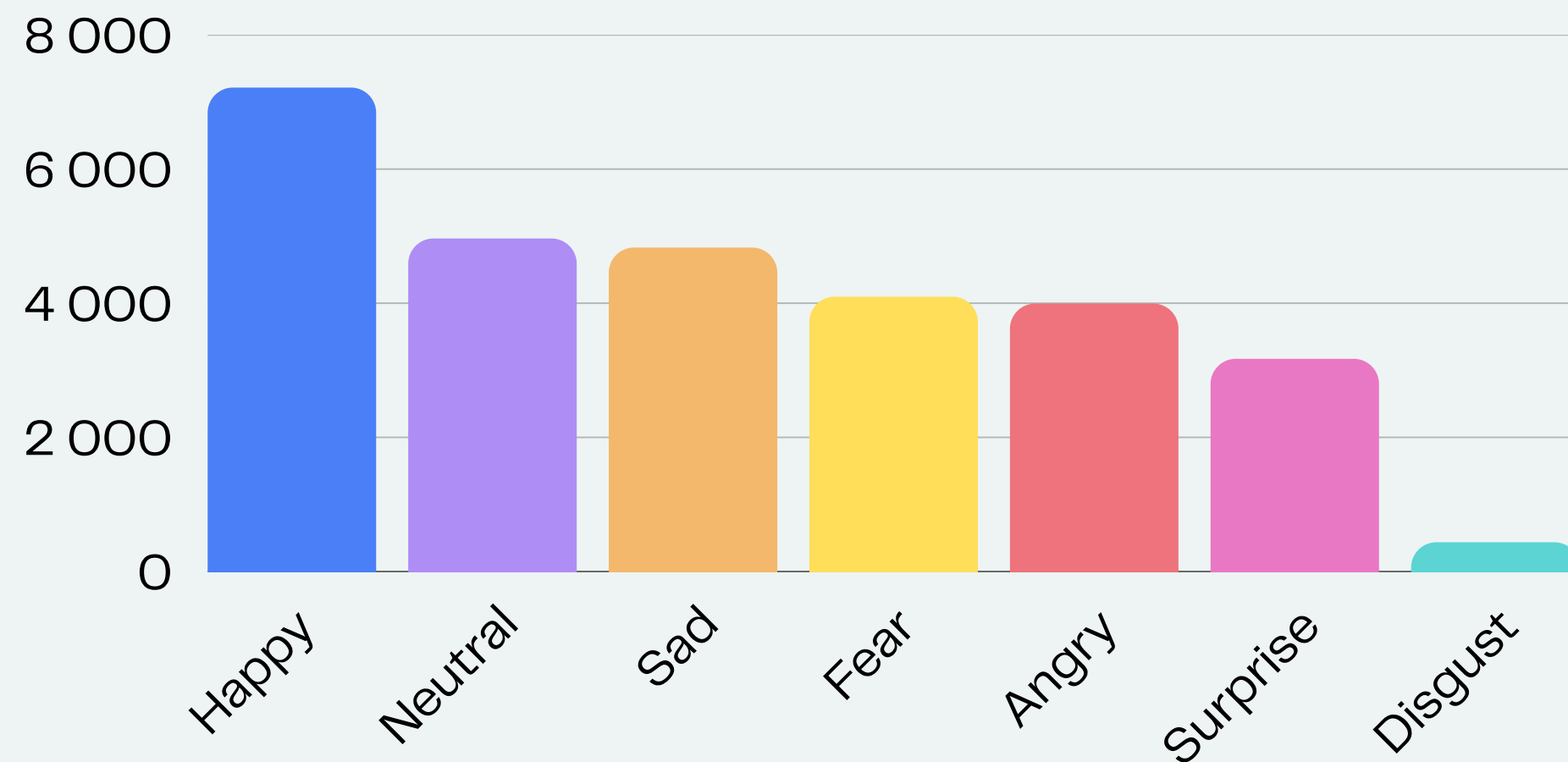
3589

VAL/TEST IMAGES

48*48

IMAGE SIZE (GRAYSCALE)

Class Distribution (training set)



Severe class imbalance:

Happy (7,215) vs Disgust (436)
— a 16.5× ratio.

Mitigation strategies:

- WeightedRandomSampler
- Inverse-frequency loss weights
- Label smoothing ($\epsilon = 0.1$)

3- METHODOLOGY

B- MODELS & OPTIMIZATION

Custom CNN

Input shape: $3 \times 96 \times 96 \rightarrow$ output: 7 logits



Each conv block = Conv +
BatchNorm + ReLU +
MaxPool

MaxPool halves spatial size at
each block

Dropout regularizes the
classifier

~1.2M trainable parameters

Why this baseline?

Baseline purpose

Train from scratch to create a fair reference before using transfer learning.

Dummy batch check

A fake batch ($2 \times 3 \times 96 \times 96$) verifies that dimensions produce 2×7 outputs.

Global average pooling

Compresses $256 \times 6 \times 6 = 9216$ features into 256 values, reducing overfitting.

Optimization

Adam, $lr=1e-3$, weight decay= $1e-4$, cosine scheduler.

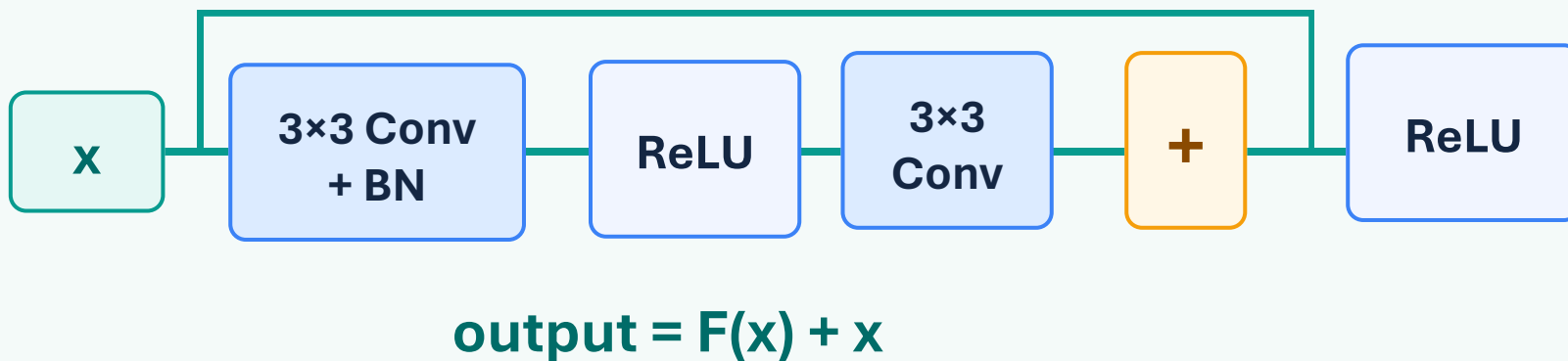
3- METHODOLOGY

B- MODELS & OPTIMIZATION

ResNet-18

Residual block idea

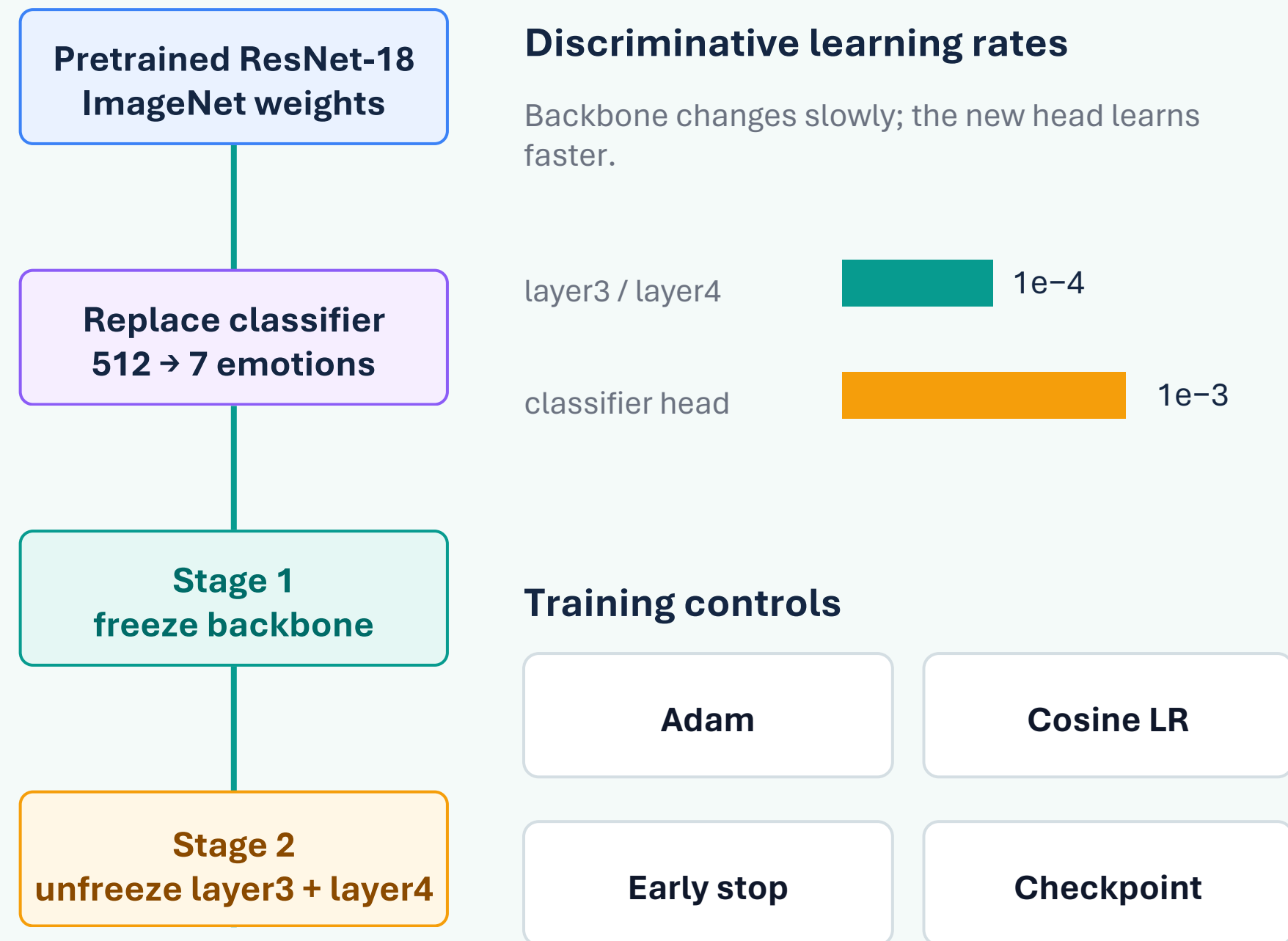
ResNet learns a residual correction $F(x)$, then adds it back to x .



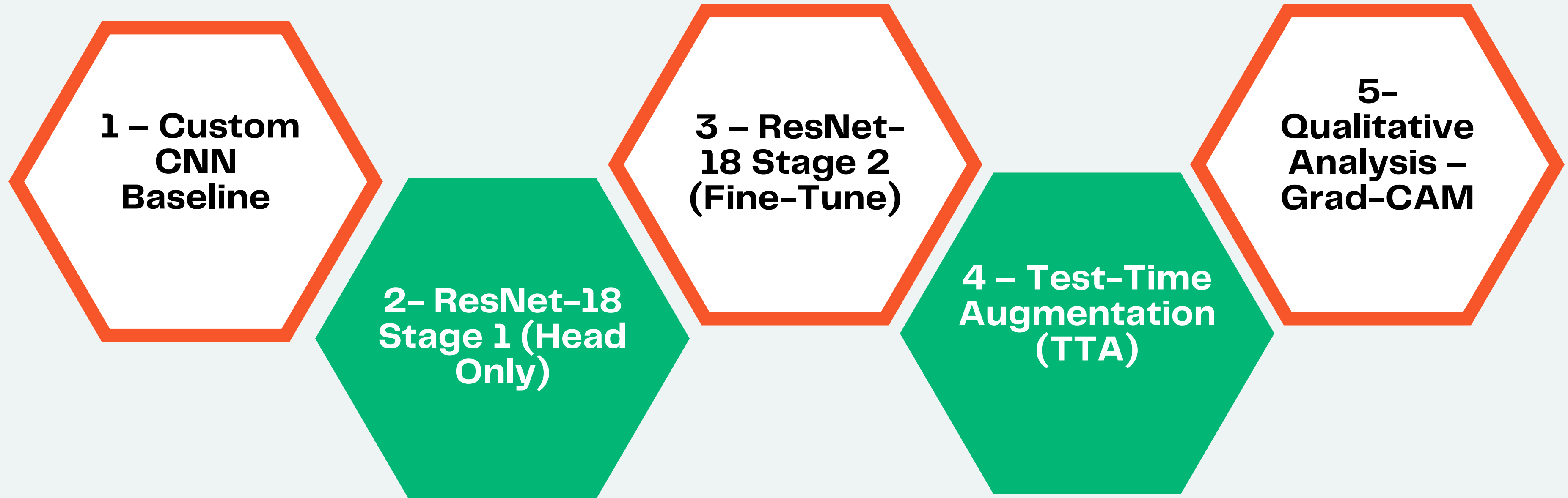
Why ResNet-18?

It reuses robust visual features from ImageNet, which is useful because FER2013 is low-resolution, noisy, and relatively small.

Transfer learning strategy



4- EXPERIMENT

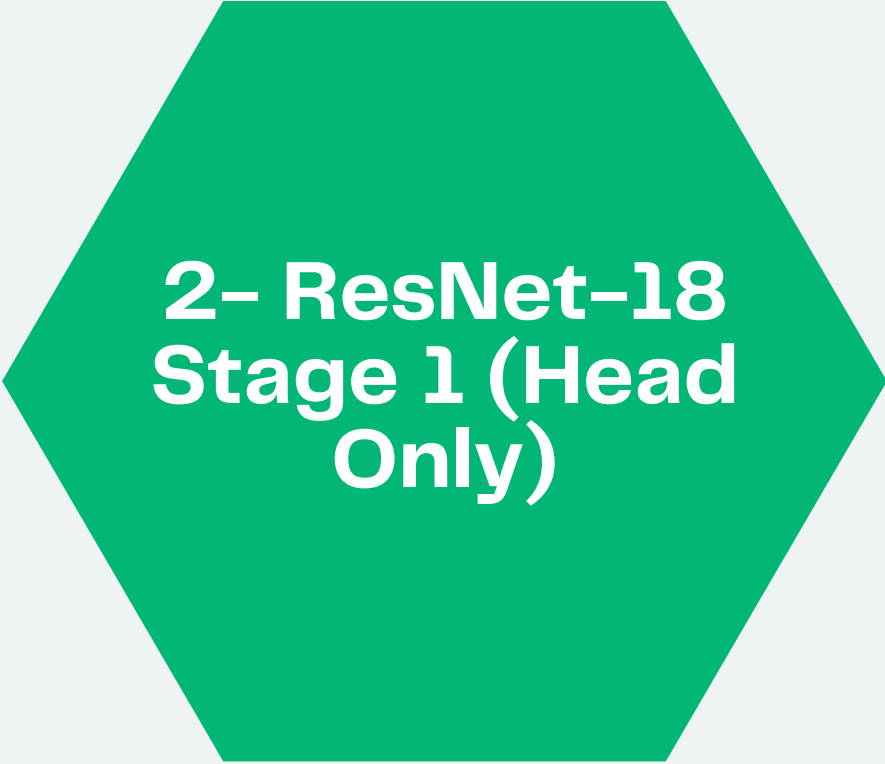


4- EXPERIMENT



1 – Custom CNN Baseline

**Trained from scratch for up to 60 epochs with early stopping.
Adam + CosineAnnealing.
Used as the lower-bound reference.**

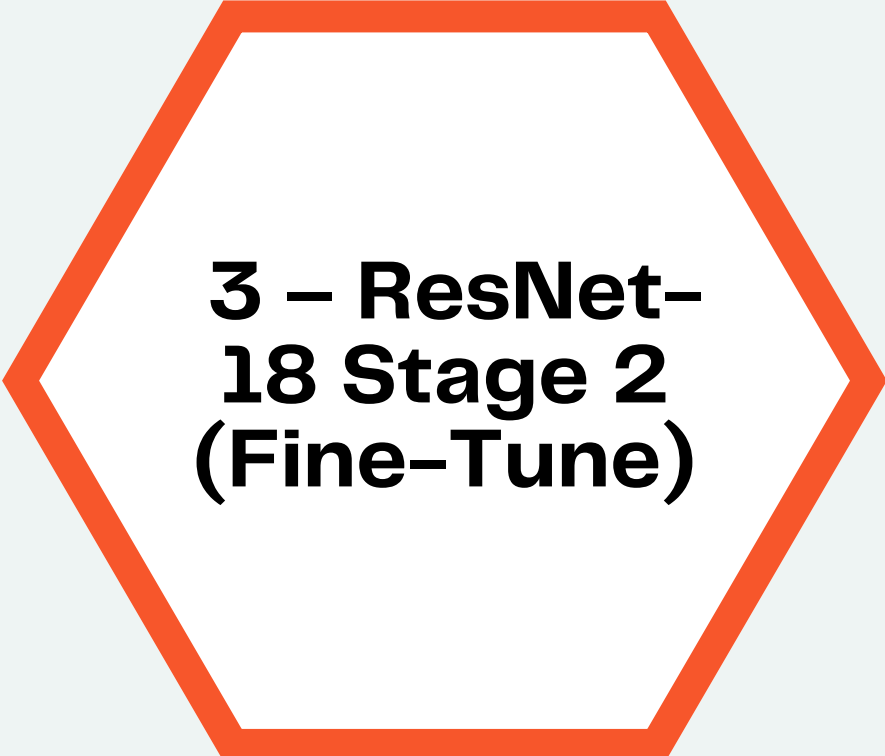


2– ResNet-18 Stage 1 (Head Only)

**Backbone frozen; only the custom FC head trained for 10 epochs.
Validates that ImageNet features transfer to grayscale emotion
data.**



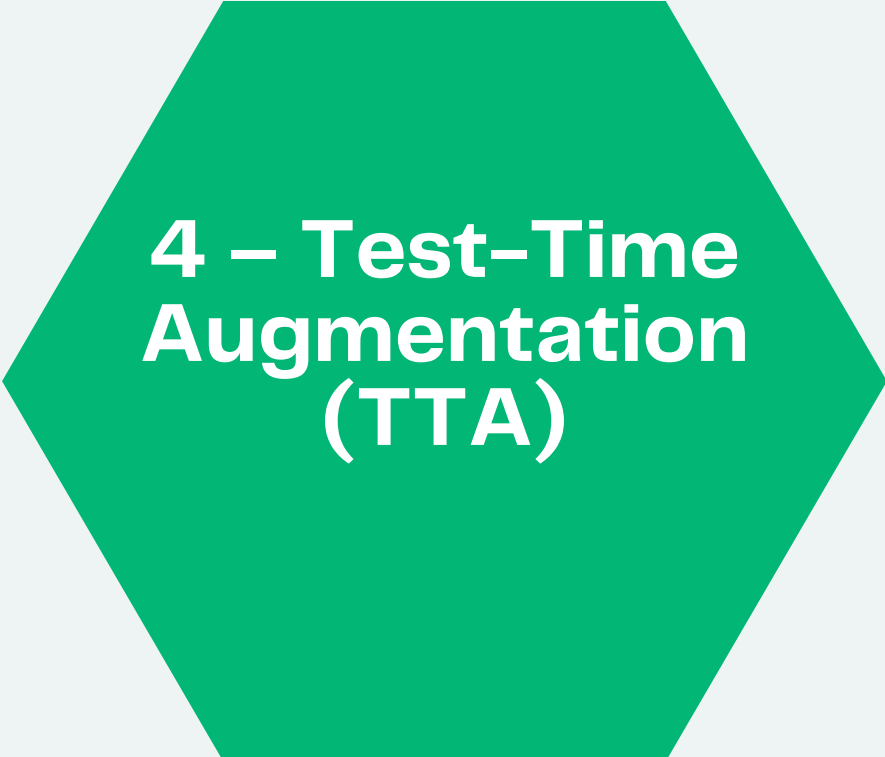
4- EXPERIMENT



3 – ResNet-18 Stage 2 (Fine-Tune)

layer3 & layer4 unfrozen with discriminative LR_s ($1e-4$ backbone / $1e-3$ head).

Allows adaptation of high-level features to facial cues.



4 – Test-Time Augmentation (TTA)

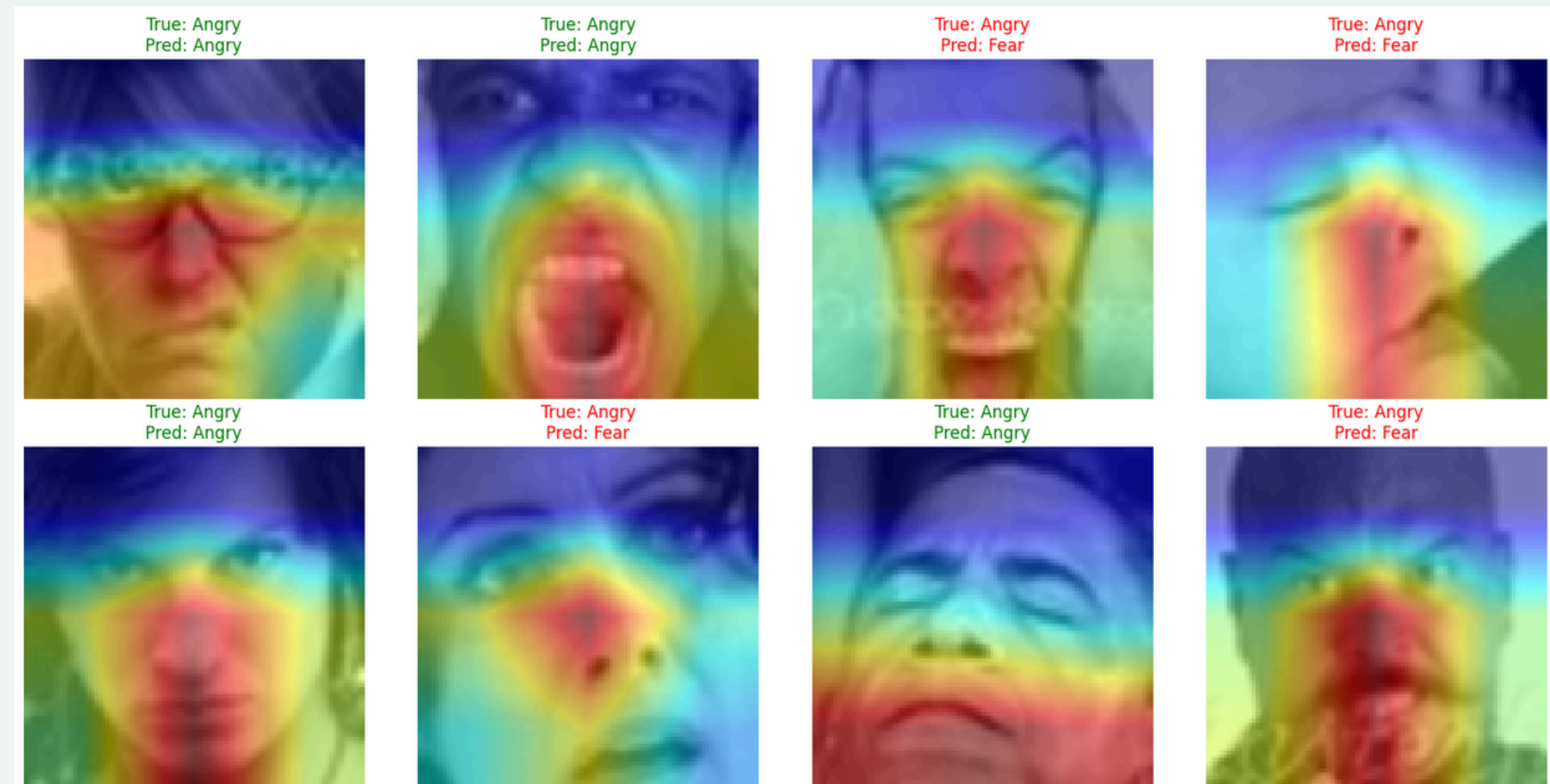
5 augmented views per image (random flips + affine shifts) averaged at inference.

Provides a final accuracy boost without retraining.

4- EXPERIMENT

5- Qualitative Analysis – Grad-CAM

Gradient-weighted Class Activation Maps from layer4 verify the network attends to discriminative facial regions (teeth for Happy, brows for Angry).



5- RESULTS

Evaluation Metrics:

Main variables tracked during training

- **train_loss / val_loss** monitor learning and overfitting
- **train_acc / val_acc** compare performance during training
- **test_acc** final performance on unseen data
- **all_preds / all_labels** used for class-level metrics

Best checkpoint selected using lowest validation loss, not the last epoch.

Precision

$$TP / (TP + FP)$$

When the model predicts an emotion, how often is it correct?

Recall

$$TP / (TP + FN)$$

Among all real examples of an emotion, how many did the model find?

F1-score

$$2 \cdot P \cdot R / (P + R)$$

Balances precision and recall, useful for imbalanced classes.

Confusion Matrix

True label × Predicted label

Shows which emotions are confused, e.g. Sad ↔ Neutral.

5- RESULTS

A- Custom CNN Training Curves

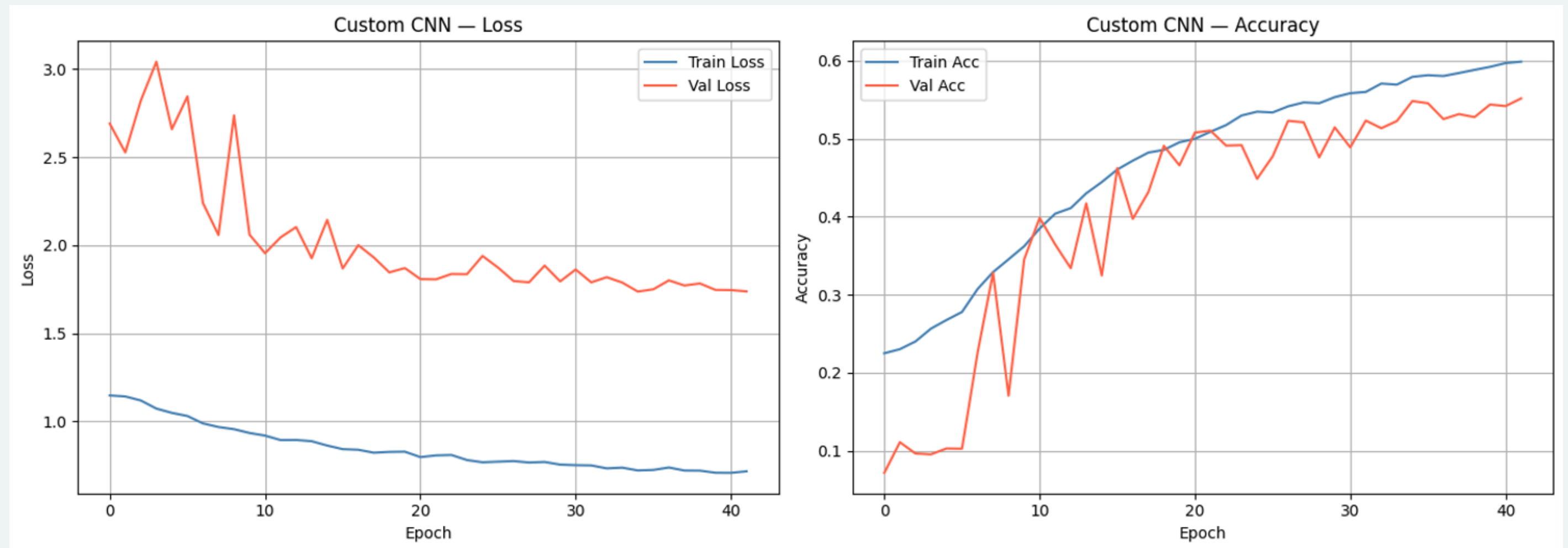


Figure 1: Custom CNN – Cross-entropy loss (left) and classification accuracy (right) on training and validation sets.

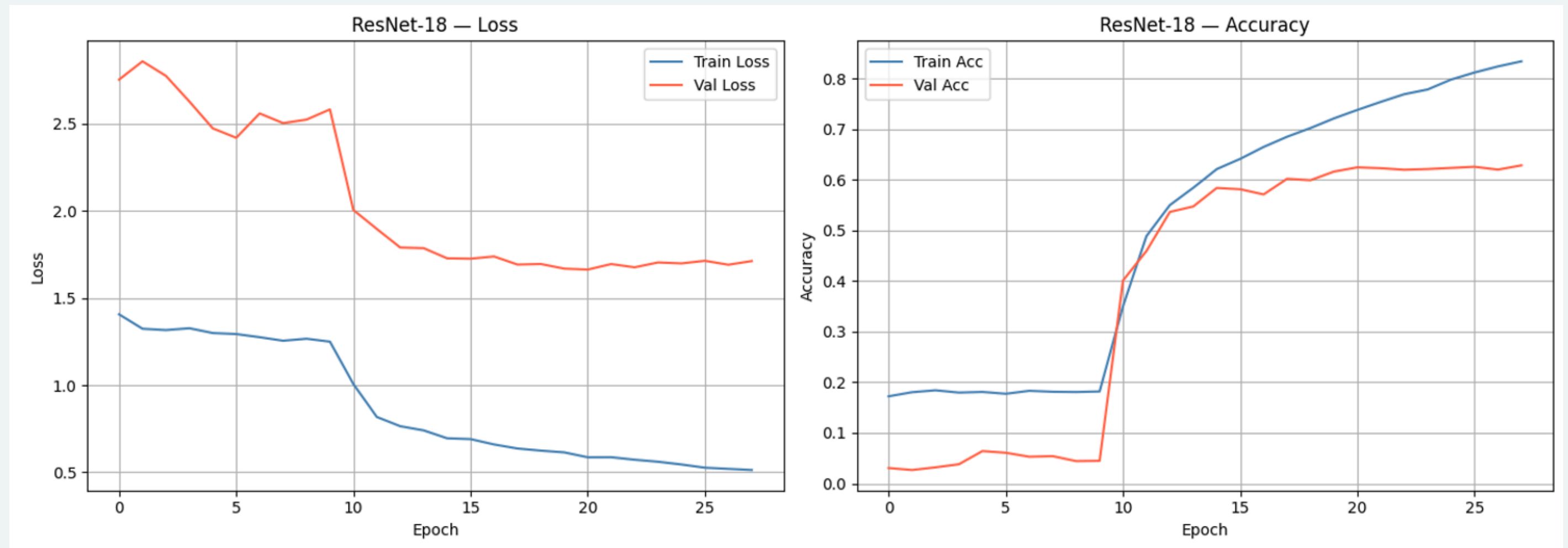
Loss and accuracy over 42 epochs (early stopping triggered).

Validation loss shows high volatility in early epochs before stabilising around epoch 15.

A widening gap between train and val accuracy in later epochs indicates overfitting.

5- RESULTS

B- ResNet Training Curves



**Figure 2: ResNet-18 – Cross-entropy loss (left) and accuracy (right).
Dashed vertical line marks the transition from Stage 1 to Stage 2 fine-tuning.**

Stage 1 (epochs 1–10): frozen backbone – minimal improvement.

Stage 2 (epochs 11–28): unfreezing layer3/layer4 causes a sharp accuracy jump. Early stopping triggers at epoch 28.

Train/val gap confirms mild overfitting.

5- RESULTS

C- ResNet Test Metrics

Class	Precision	Recall	F1-score	Support
<i>Angry</i>	0.56	0.52	0.54	491
<i>Disgust</i>	0.44	0.84	0.57	55
<i>Fear</i>	0.49	0.50	0.50	528
<i>Happy</i>	0.89	0.80	0.84	879
<i>Neutral</i>	0.62	0.62	0.62	626
<i>Sad</i>	0.47	0.50	0.48	594
<i>Surprise</i>	0.74	0.75	0.75	416
Accuracy			0.63	3589
Macro avg	0.60	0.65	0.62	3589
Weighted avg	0.64	0.63	0.64	3589

5- RESULTS

D- Confusion Matrix (ResNet-18, Test Set)

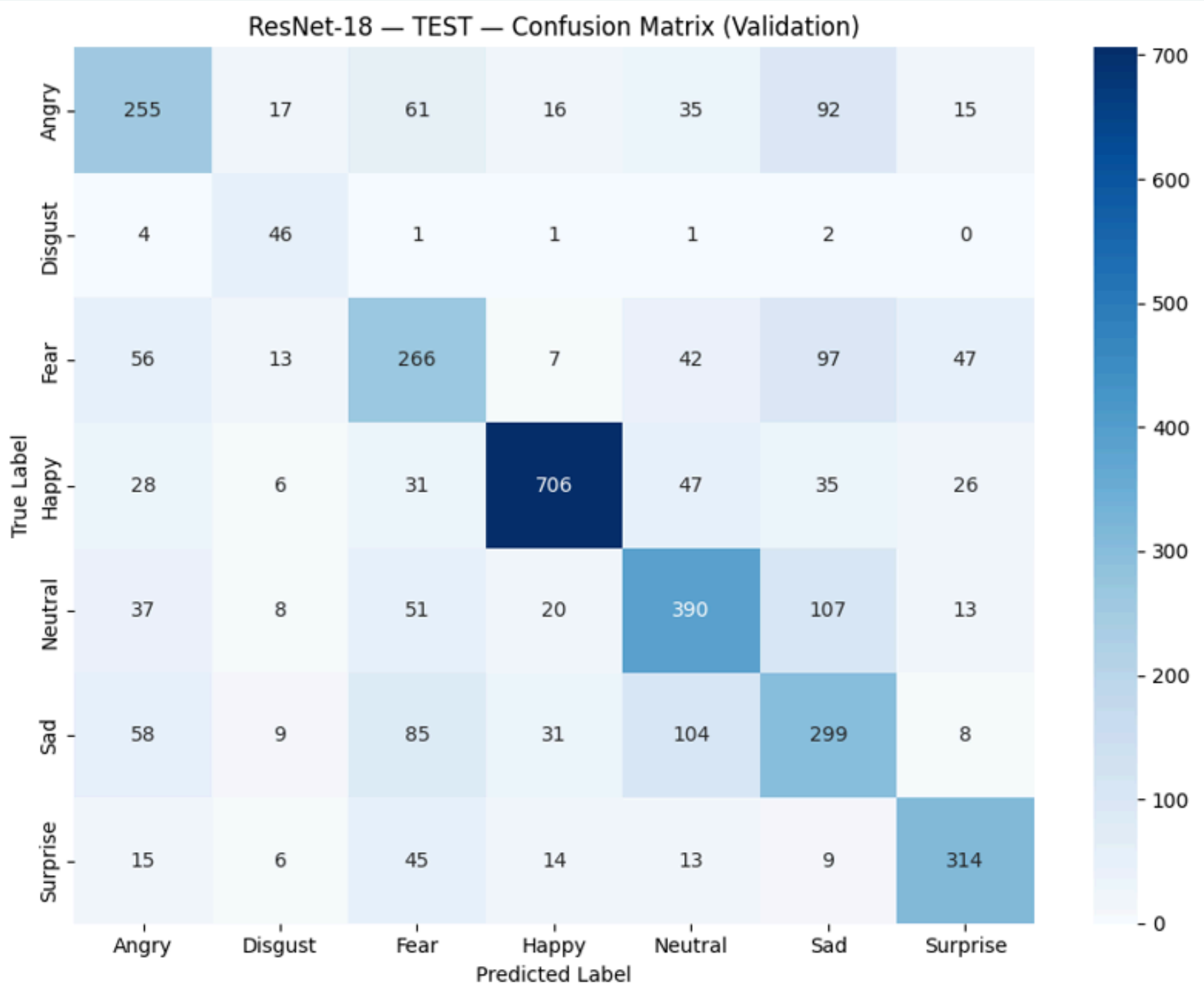


Figure 3: Confusion matrix for ResNet-18 on the test set.
Rows = true labels, columns = predicted labels.

Key observations

Happy & Surprise

Highest diagonal counts — visually distinct cues (teeth, raised eyebrows).

Sad ↔ Fear

97 Fear predicted as Sad; 85 Sad predicted as Fear — highest confusion pair.

Sad → Neutral

100 Sad faces predicted as Neutral — subtle expression overlap.

Disgust

Small class (100 test samples) but 62 correct — high recall thanks to loss weighting.

Angry

Confused with Sad (60) and Neutral (55) — shared furrowed-brow features.

6- DISCUSSION

A- What worked well

Transfer Learning: ResNet-18 leveraged ImageNet features to jump 6 pp. over the from-scratch CNN.

Staged Fine-Tuning: Transfer Learning: ResNet-18 leveraged ImageNet features to jump 6 pp. over the from-scratch CNN.

Imbalance Mitigation: Combining WeightedRandomSampler + loss weights significantly improved minority-class recall (especially Disgust).

Grad-CAM: Heatmaps confirmed the model attends to genuine facial features, not background artifacts.

B- Limitations & Challenges

Negative Emotion Confusion: Fear, Sad and Angry share visual features — confusion matrix shows systematic misclassification chains between these classes.

Overfitting: Train accuracy consistently pulled ahead of val in late epochs despite dropout, L2 regularisation, and label smoothing..

Dataset Noise: FER2013 images are scraped from the web and inconsistently labelled — this creates a hard ceiling on achievable accuracy.

Architecture Scope: No attention-based or transformer-based models (ViT, DeiT) were explored — these currently dominate the SOTA.

7- CONCLUSION

Main Finding:

ResNet-18 fine-tuning (63.4%) substantially outperforms a from-scratch CNN (57.1%) on FER2013, approaching the human-level ceiling of ~65%.

Robstness:

Dual imbalance handling (sampler + loss weights + label smoothing) was essential to prevent collapse on minority classes like Disgust.

Future Work:

Vision Transformers (ViT, DeiT), larger datasets (AffectNet), and multi-modal fusion (face + body pose) could push well beyond 65%.

Key Technique:

Discriminative LRs + staged unfreezing proved crucial — freezing the backbone first prevented overwriting general visual features too early.

Interpretability:

Grad-CAM confirmed the model reasons about genuine facial structures, not spurious background correlations.

THANK YOU!

Questions?

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